Safety systems · United Kingdom

British colour-light signals show **how many sections ahead are clear**, not the speed: green, double yellow, single yellow, red. Speed comes from the **lineside speed signs** and the route knowledge. Junction indicators and route indicators show a diverging route. **AWS** warns you at every caution, **TPWS** stops you at a red signal or an overspeed, the **DSD / vigilance** checks that you are awake. Quadrant controls are from TheJAG's UK profiles.

Safety systems

AWS
Automatic Warning System

A magnet 180 m before colour-light and distant signals and warning boards. **Bell / chime** = clear (green). **Horn** = caution, red or a speed warning: acknowledge within about **2.7 s** or the brakes apply. The black and yellow **sunflower** then shows you did.

WHERE Every Network Rail route

QUADRANT **Gear lever** = AWS reset; left aux square = AWS (/ TPWS) isolation

TPWS
Train Protection and Warning System

Loops at red signals, speed restrictions and buffer stops. **Train stop**: passing a red signal brakes the train. **Overspeed sensor**: approaching too fast brakes it. Brake demand lamp flashes; the brakes release 60 s after the demand.

WHERE Every Network Rail route

QUADRANT Shares the AWS isolation (left aux square) on most units; Train Stop Override in the cab

DSD / vigilance
Driver's Safety Device, Driver Vigilance Device (DVD)

A pedal or handle held while driving; let go and the train brakes. Vigilance adds an alarm about every 60 s that you acknowledge.

WHERE Every modern unit and loco

QUADRANT Gear lever acknowledges in TSW (0); right aux square = DSD / DVD isolation on most units

DRA
Driver's Reminder Appliance

A button you set when stopped at a red signal or a platform with the signal at danger: it cuts traction until you release it.

WHERE Units and locos that have one

QUADRANT Round aux on the 385 and the 90; in the cab on the others

Tripcock
London Underground trainstops

A trainstop arm next to each LU signal is raised at red; passing it trips the tripcock on the train and brakes it.

WHERE LU 1938 and 1972 Stock (Bakerloo line)

QUADRANT 1972: **ENG 1** tripcock reset, gear lever valve reset

TVM 430 / ETCS
Cab signalling

HS1 uses French TVM 430 cab signals (no lineside signals) KVB in the St Pancras area (not simulated). No TSW 7 UK route in this collection has ETCS.

WHERE Southeastern Highspeed (Class 395 on HS1)

Colour-light signals (4-aspect)

Rays = flashing. A 3-aspect signal has no top yellow and no double yellow.



Green
Proceed; the next signal shows a proceed aspect.

DRIVE Line speed. AWS bell.



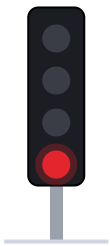
Double yellow
Preliminary caution: the next signal shows single yellow.

DRIVE AWS horn: acknowledge. Start to brake.



Single yellow
Caution: be prepared to stop at the next signal.

DRIVE Acknowledge AWS; brake to stop at the next signal.



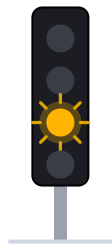
Red
Danger: stop.

DRIVE Stop before the signal; TPWS train stop.



Flashing double yellow
Diverging junction ahead at a speed; next signal flashing single yellow.

DRIVE Acknowledge; brake toward the junction speed.



Flashing single yellow
Next signal shows a junction indicator for a diverging route.

DRIVE Be at the turnout speed at the junction.

Sources: RSSB GERT8000 Rule Book (modules S1, S3, TW1) · RSSB GKRT0045 lineside signals · Network Rail · en.wikipedia: UK railway signalling, AWS, TPWS, Driver's Safety Device, DRA

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Route, subsidiary and ground signals



Junction indicator
Five white lights: the route diverges to the left (or right).

DRIVE Take the turnout at its speed sign.



Route indicator
Theatre box: a lit letter or number names the route or platform.

DRIVE Check the route is the one you expect.



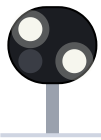
Position light subsidiary
Red with two white lights: pass the red signal; the line ahead may be occupied.

DRIVE Pass at caution, ready to stop short of any obstruction.



Ground signal: stop
Two reds horizontal (older signals: red and white): stop.

DRIVE Stop short of it.

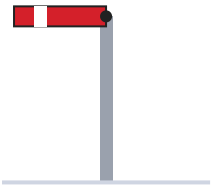


Ground signal: proceed
Two white lights at 45°: proceed (shunt).

DRIVE Proceed at caution to the next signal or limit.

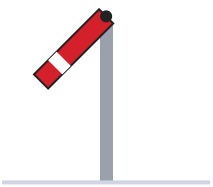
Semaphores

Stop arm: red with a white stripe. Distant arm: yellow with a black chevron and fishtail end. Lower quadrant arms drop for off, upper quadrant arms rise.



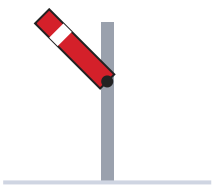
Stop: on
Arm horizontal: danger.

DRIVE Stop before the signal.



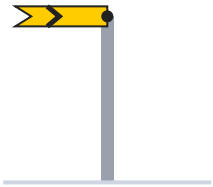
Stop: off (lower quadrant)
Arm lowered: proceed.

DRIVE Proceed.



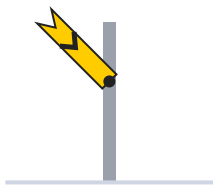
Stop: off (upper quadrant)
Arm raised: proceed.

DRIVE Proceed.



Distant: on
Caution: the stop signal ahead may be at danger.

DRIVE AWS horn: acknowledge, be ready to stop.



Distant: off
All the stop signals of the box ahead are off.

DRIVE AWS bell; proceed.

Speed signs and boards



Speed sign
Permissible speed from here, in mph (older lines: yellow cut-out figures).

DRIVE Lower: by the sign. Higher: once the rear has passed.



PSR warning
A lower permissible speed ahead; AWS magnet 180 m before it.

DRIVE AWS horn: acknowledge, brake to the speed.



TSR warning
Temporary restriction ahead; AWS magnet. Two white discs (flashing white lights on lit boards).

DRIVE AWS horn: acknowledge, brake to the speed.



TSR speed
Temporary limit starts here.

DRIVE Be at the speed by the board.



TSR end
Temporary limit ends.

DRIVE Accelerate once the rear has passed.



AWS cancelling
An AWS magnet for the other direction: no warning for you. Yellow with a black cross at a TSR.

DRIVE Ignore any AWS indication here.

Sources: RSSB GERT8000 Rule Book (modules S1, S3, TW1) · RSSB GKRT0045 lineside signals · Network Rail · en.wikipedia: UK railway signalling, AWS, TPWS, Driver's Safety Device, DRA

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AWS and TPWS at a glance

Event	In the cab	What to do
Green signal, distant off	Bell or chime; sunflower black	Nothing
Caution, red, PSR / TSR warning	Horn	**AWS reset** (gear lever) within about 2.7 s; sunflower turns yellow and black
AWS not acknowledged	Brakes apply	Stop, acknowledge, then release the brake
TPWS overspeed or train stop	Brake demand lamp flashes, brakes apply	Acknowledge (lamp goes steady); report to the signaller; the brakes release 60 s after the demand
DSD released / vigilance alarm missed	Buzzer, then brakes	Stop, hold the DSD / acknowledge, release the brake

How to drive under AWS, TPWS and the DSD

Approaching a caution

1. Double yellow: AWS horn, press **AWS reset** (gear lever). Start braking.
2. Single yellow: AWS horn again, reset. Brake to stop at the next signal.
3. Keep below the TPWS trigger speed on the approach to the red; stop short.

At a red signal

1. Stop before the signal. Set the **DRA** if you are waiting.
2. Contact the signaller if the signal does not clear.
3. To pass at danger with authority, press **Train Stop Override** (in the cab): it covers the train stop for about 20 s (60 s on the 66), never the overspeed sensor.
4. Release the DRA when the signal clears, then take power.

TPWS intervention

1. Brakes apply and the brake demand lamp flashes.
2. Leave the controls; let the train stop.
3. Acknowledge with **AWS reset**: the lamp goes steady. Report to the signaller; the brakes release 60 s after the demand.

Speed restrictions

- PSR and TSR warning boards carry an AWS magnet: horn, reset, brake.
- Be at the speed by the speed sign; TPWS overspeed sensors guard some restrictions.
- Higher speed only once the rear of the train has passed the sign.

DSD and vigilance

1. Keep the DSD / DVD switched in (right aux square: isolation off).
2. When the vigilance buzzer sounds, acknowledge with the **gear lever**.
3. Missed: the brakes apply; stop, acknowledge, release and drive on.

London Underground (1938, 1972 Stock)

- Signals: red / green; a trainstop arm is raised at red.
- Passing a red trips the tripcock: emergency brake.
- A trip counts as a signal passed at danger: stop, report, then reset the tripcock (**ENG 1**) and the valve (**gear lever**, 1972) and continue slowly.

Before departure: AWS / TPWS and DSD isolation lights out: on most units the aux squares are isolation switches, unlit = in service (the 380 starts isolated). Flying Scotsman: left aux lit = AWS on; the LMS steam locos have no AWS. DRA released, the signal off.

Sources: RSSB GERT8000 Rule Book (modules S1, S3, TW1) · RSSB GKRT0045 lineside signals · Network Rail · en.wikipedia: UK railway signalling, AWS, TPWS, Driver's Safety Device, DRA

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